Tianrui Gu

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EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree Class of 2029
Applied Mathematics and Computational Sciences — Computer Science Track Kunshan, China
Relevant coursework: MATH 201, 202, 203, 206, 305; COMPSCI 101, 201. Spring 2027: MATH 405, COMPSCI 308, 205, MATH 302.

RESEARCH INTERESTS

Mathematical foundations of machine learning: harmonic analysis, geometry, optimization, and probability for understanding and improving learned representations, with applications to medical imaging.

RESEARCH EXPERIENCE

Password Guessing Based on Deep Learning — Yuan Pei Young Scholars Program, Peking University Jun – Aug 2024
Advisor: Prof. Lianghuan Kang, Peking University Beijing, China

- Built a WGAN-GP password-guessing model on the RockYou dataset, extending PassGAN with an additional generator linear block and batch normalization before each ReLU in the residual blocks.
- Trained and evaluated the model in Google Colab; compared hit rates with the original PassGAN/WGAN-GP baseline and analyzed how password length and letter–digit composition affect guessing success.
- Identified early stopping as a practical training strategy from discriminator/generator loss minima; translated findings into user-facing security recommendations (≥15 characters, mixed character classes, no reuse, 2FA).

Paper: View PDF

SELECTED WRITING

"Mathematical Notes I: Linear Algebra and the Spectral Theorem" (link); "Mathematical Notes II: The Spectral Theorem, Differential Equations, and Laplace Transforms" (link). Both notes are in Chinese.

INTERNSHIP

GEEX Finance (Shanghai GEEX Intelligence Technology Group Co., Ltd.) Feb – Apr 2025; Nov 2025 – Present
Intern, Consumer Finance Shanghai / Shenzhen, China

- Designed model algorithms and ran effect testing for a merchant anti-fraud model built on graph neural networks, targeting merchant default risk in scenario-based consumer lending.
- Supported licensing negotiations for a mobile-phone installment business in the DRC with the Honorary Consul of Norway; sourced handset inventory in Huaqiangbei and scoped category expansion.

LEADERSHIP & SERVICE

President, DKU Computer Science Club Jan 2026 – Present

- Lead the club's executive team and year-round programming; built and maintain www.dkucs.com, wiki.dkucs.com, and the HackDKU 2026 site using Next.js, React, Tailwind CSS, and MDX.
- Hosted HackDKU 2026, a week-long Ethics & AI hackathon with four tracks (Health, Finance, Education, Luxshare); contributed to sponsorship outreach (GEEX Finance, Caidazi, Zhongke Jincai) and event operations.
- Organize "AI = Amplified Individuals" (在 AI 中，看见具体的人), the DKU edition of a WAIC Shanghai forum with four themed sessions and ~200 expected attendees (Nov 2026).
- Building a succession plan for the next HackDKU team.

Problem Designer, DKU Math Union Integration Bee 2026 2026

- Contributed problems and full solutions for integral computation, Fourier analysis, Laplace transforms, and multivariable vector calculus; event format: 20-minute qualifier + 16-player bracket, 77 problems across five rounds. Problem set: View PDF

TECHNICAL SKILLS

Programming: Python, Java, LaTeX, SQL (basic) | ML: PyTorch, NumPy | Web: Next.js, React, Tailwind CSS, MDX | Tools: Git, Google Colab, Jupyter
Languages: Mandarin (native), English (fluent)